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Deep learning convolutional neural networks for morphological identification of field-caught *Anopheles stephensi* in Ethiopia, using smartphone images of specimens

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Abstract

Background *Anopheles stephensi*, an invasive malaria vector historically confined to South Asia and the Arabian Peninsula, has established reproducing populations across the Horn of Africa and continues to expand its range. Its adaptability to a range of environments, with its impact described most prominently in urban and peri-urban settings, and morphological similarity to native African anophelines, particularly *An. gambiae* s.l., makes reliable field identification difficult by field entomologists across much of Africa who have had limited prior exposure to the species. Existing molecular methods are impractical for routine field use, creating an urgent need for scalable, accessible identification tools to support surveillance programs responding to this emerging threat.

Methods Three deep learning convolutional neural network (CNN) classifiers were developed and validated for detecting *An. stephensi* from photographic images of mosquito specimens collected from sentinel sites across Ethiopia and Uganda, verified for species identity by PCR or morphological identification. VectorCam, a standardized field imaging device, served as the primary imaging platform, with a supplementary handheld smartphone dataset collected using clip-on macro lenses (10X and 15X). An EfficientNet-B1 architecture was fine-tuned using transfer learning from a pretraining dataset of over 60,000 labeled photographs of field-collected mosquito specimens from Uganda. Three classifiers were developed: a VectorCam multiclass classifier distinguishing seven species classes including *An. stephensi*, a VectorCam binary classifier distinguishing *An. stephensi* from all other species, and a handheld smartphone binary classifier for photographic images taken using a handheld smartphone. All models were evaluated using 5-fold cross-validation and a majority voting ensemble of the trained classifiers.

Results Against PCR and morphologically verified ground-truth identifications, the VectorCam multiclass classifier achieved an *An. stephensi* sensitivity of 98.74% and overall accuracy of 96.02%. The VectorCam binary classifier achieved an *An. stephensi* sensitivity of 97.48% and a specificity of 99.25%. The handheld binary classifier achieved an *An. stephensi* sensitivity of 91.09% and a specificity of 84.40%. Gradient-weighted Class Activation Mapping confirmed that classification decisions were driven by wing venation and leg morphology, consistent with established entomological identification criteria.

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Conclusions Deep learning image classification using VectorCam achieves high accuracy for field detection of *An. stephensi*, with the binary classifier demonstrating performance suitable for operational vector surveillance, providing a practical, scalable solution for decentralized *An. stephensi* surveillance across malaria-endemic regions of Africa, reducing reliance on molecular verification or specialist taxonomic training for field-level species identification.

Keywords *Anopheles stephensi*, Invasive malaria vector, Deep learning, Convolutional neural network, VectorCam, Mosquito species identification, Vector surveillance, East Africa

Background

The invasion of *Anopheles stephensi* into sub-Saharan Africa represents a critical and emerging threat to malaria control efforts across the continent. This species, historically confined to South Asia and the Arabian Peninsula, has established reproducing populations across the Horn of Africa, with its geographic range continuing to expand [1–3]. The behavioral and epidemiological profile of *An. stephensi* differs markedly from that of the dominant native vectors that existing surveillance and control programs have been designed to address. Unlike native African anophelines that primarily breed in rural natural water bodies, *An. stephensi* is highly adaptable to a range of environments. Its impact has been described most prominently in urban and peri-urban settings, where it breeds readily in artificial water storage containers such as tanks, cisterns, and other man-made receptacles, similar to that of Aedes aegypti [2, 4]. As a result, existing vector surveillance infrastructure, which has largely been optimized around rural *An. gambiae* s.l. and *An. funestus* s.l. populations, is poorly suited to detect or respond effectively to the presence and expansion of this invasive species [5]. The presence and establishment of *An. stephensi* in urban centers therefore poses a compounding challenge for malaria control programs that have made significant progress in reducing transmission in traditionally rural settings [2, 6].

A fundamental prerequisite for effective surveillance and control of *An. stephensi* is reliable species-level identification in the field. However, this presents a substantive and practical challenge. The species shares key morphological characteristics, including body size, coloration, and wing patterning, with native East African anophelines, particularly An. gambiae s.l. (Fig. 1), which is among the most important malaria vectors in Sub-Saharan Africa [6, 7]. These overlapping traits make accurate visual identification difficult, especially in field settings where rapid and confident differentiation between species is critical for guiding surveillance and control responses. As *An. stephensi* has only recently established in the region [1, 2], field entomologists across much of Africa have had limited prior exposure to the species, further compounding this challenge. Morphological identification of mosquito species has long been a cornerstone of vector surveillance programs, providing crucial data for understanding species distributions and bionomic traits, and consequently driving targeted control strategies [8]. However, distinguishing An. stephensi from closely related native species based on morphological characteristics alone is expertise-intensive, time-consuming, and prone to errors [9]. Gold-standard molecular methods such as PCR provide definitive identification but require significant monetary investments in laboratory infrastructure and trained personnel that preclude their use for routine, real-time surveillance in field settings. The consequences of misidentification are significant: flawed estimates of the species' invasion range, misdirected control interventions, and delayed public health responses, all in the face of malaria outbreaks and increasing malaria transmission.

Artificial intelligence (AI)-based image classification has shown considerable promise as an approach to address this identification bottleneck at scale. Convolutional neural networks (CNNs) are a class of deep

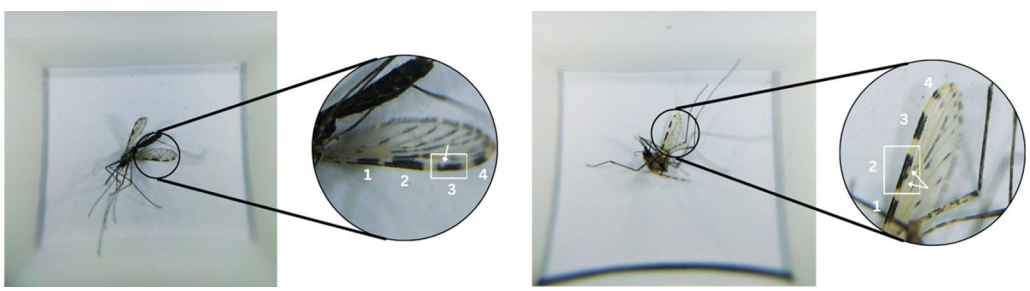


Fig. 1 Comparative wing venation patterns of *Anopheles gambiae* (left) and *Anopheles stephensi* (right), captured using VectorCam

learning models that process images through successive layers of filters, each extracting visual features at increasing levels of abstraction. Early layers detect low-level features such as edges and textures, while deeper layers detect patterns that correspond to class-level characteristics. By training on labeled image datasets, CNNs learn to associate these visual patterns with class labels. CNN models trained on mosquito images have achieved high accuracy in species-level classification across a number of prior studies [10–12], with recent work also demonstrating field deployment feasibility of such approaches [13]. However, achieving reliable performance under genuine resource-limited field conditions, where lighting is variable, operators may lack specialist training, and infrastructure is minimal, remains an open and important challenge for the field. VectorCam is a purpose-built, low-cost field imaging device developed specifically to address these constraints in vector surveillance settings (Fig. 2) [11, 13]. The device integrates a standardized optical setup with a guided specimen placement interface and onboard AI processing, enabling it to produce consistent, high-resolution images of mosquito specimens and display species identifications in real time without requiring internet connectivity. Its onboard CNN model was pretrained on over 60,000 labeled mosquito images collected from Uganda, covering multiple *Anopheles*, *Culex*, and *Mansonia* species, and has been validated in vector surveillance field settings across East Africa [11]. By standardizing image acquisition, VectorCam eliminates the operator-dependent variability in image quality that represents the primary limitation of smartphone-based imaging approaches, while maintaining a form factor that can be operated by non-specialist surveillance staff in the field.

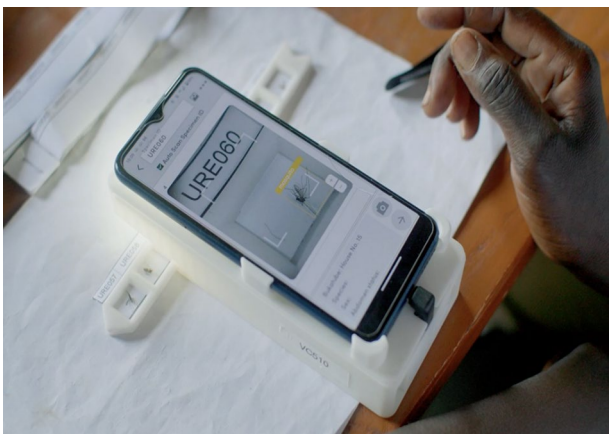


Fig. 2 VectorCam field imaging device in operation at a sentinel surveillance site in Uganda

In this study, we present a deep learning-based system for detecting *An. stephensi* from field-collected specimens, with VectorCam serving as the primary deployment platform. Mosquito specimens collected from sentinel sites across Ethiopia, one of the first African countries to document established *An. stephensi* populations, were molecularly or morphologically verified for species identity and used to train and validate CNN classifiers capable of distinguishing *An. stephensi* from co-occurring East African mosquito species including *An. gambiae* s.l., *An. funestus* s.l., *Culex*, and *Mansonia*. To extend accessibility to surveillance programs that do not have access to VectorCam devices, a supplementary web application was additionally developed to support species identification from handheld smartphone images captured with clip-on macro lenses. Together, these two platforms provide a flexible, tiered surveillance capability that can be adapted to different resource environments across the continent.

Numbered markers indicate key morphological features used to distinguish the two species. On the wing, markers indicate the pale spot interruptions in the second main dark area of vein 1. *Anopheles gambiae s.l.* has one pale spot in this region, while *An. stephensi* has two pale spots.

Methods

Three deep learning CNN classifiers were developed and validated for detecting *An. stephensi* among co-occurring mosquito species in East Africa. Two imaging modalities were evaluated: VectorCam as the primary platform, and a handheld smartphone with clip-on macro lens as a supplementary approach. VectorCam was selected as the primary platform as it can augment the *An. stephensi* detection capabilities of surveillance programs that already employ a standardized field imaging system, while the handheld smartphone approach was developed to extend detection capability to programs that have access only to smartphones and widely available clip-on macro lenses (10X, 15X), enabling them to incorporate *An. stephensi* identification into their existing entomological surveillance workflows without the need for dedicated imaging hardware. All specimens used for model training and validation were field-caught and verified for species identity prior to inclusion, either by PCR (*An. stephensi*, *An. gambiae* s.l., *An. funestus* s.l., *Culex*, *Mansonia*) or by morphological identification by a trained entomologist under direct microscopic visualization (*An. other*, non-mosquitoes). Specimens were excluded from the dataset only in cases where a definitive ground-truth species identity could not be established, for example where a specimen was not viable for PCR amplification or did not meet the morphological criteria required for

confident identification. Specimens were also excluded where the corresponding image did not meet minimum quality requirements for model training, for example where the image was blurry, did not contain a visible specimen, contained more than one specimen within the same frame, or captured only a partial view of a specimen rather than the complete specimen. A specific exclusion rate was not tracked for either category, as such specimens were identified and removed prior to their incorporation into the study dataset at the point of collection. The specimen counts reported in Table 1 therefore reflect the complete and final set of specimens with confirmed ground-truth identifications used for model development.

Species identification methods

1. Morphological Identification

Specimens identified morphologically were examined by trained entomologists under direct microscopic visualization using established taxonomic keys [14]. Morphological identification was applied to *An. other* and non-mosquito specimen classes, as well as to a subset of *An. gambiae* s.l. and *An. funestus* s.l. specimens. All *An. stephensi* specimens were exclusively verified by PCR prior to inclusion in the dataset; no *An. stephensi* specimens identified by morphology alone were included in model training or validation.

2. Molecular Identification (PCR)

PCR verification was performed at the University of Notre Dame (Notre Dame, IN, USA) using established molecular protocols [15–17]. Specimens submitted for PCR verification included all *An. stephensi* specimens, all *Culex* and *Mansonia* specimens, as well as a subset of *An. gambiae* s.l. and *An. funestus* s.l. specimens.

Data collection and preparation

Mosquito specimens used for algorithm development were obtained from existing entomological surveillance and research activities conducted in Ethiopia and Uganda. Specimens were collected by collaborating programs and studies using routine mosquito collection methods, including pyrethrum spray catches (PSC) and CDC light traps (CDC-LT), and were subsequently provided to the study team for imaging and model development. Because specimens were contributed from multiple pre-existing studies and surveillance activities, specimen-level metadata on the original collection method were not consistently available, and the number of specimens collected by each trapping method could therefore not be determined.

Anopheles stephensi specimens were obtained from surveillance sites in Ethiopia. *Anopheles gambiae* s.l. specimens were obtained from both Ethiopia and Uganda, while *An. funestus* s.l., *Culex*, *Mansonia*, and other *Anopheles* specimens were obtained from Uganda.

VectorCam imaging

VectorCam was used as the primary imaging platform in this study (Fig. 2) [11, 13]. The device provides standardized image acquisition through a fixed optical setup, controlled illumination, and a guided specimen placement interface, ensuring high reproducibility of image quality across different operators and field sites. This standardization is particularly important for maximizing classification accuracy, as it minimizes the operator-dependent variability in lighting and magnification that typically limits the consistency of smartphone-based imaging. Specimen orientation and position within the imaging field were intentionally allowed to vary during image acquisition to increase positional diversity in the training dataset.

In total, 4,763 VectorCam images were captured across all species classes and used for model training, validation, and testing. All specimens were imaged fresh, typically

Table 1 Number of specimens per species class, by verification method and imaging platform

Species	VectorCam-PCR verified	VectorCam-Morphology verified	Handheld-PCR verified	Handheld-Morphology verified
An. stephensi	791	0	396	151
An. gambiae s.l	656	414	359	143
An. funestus s.l	395	426	–	–
Culex	240	336	–	–
Mansonia	233	350	–	–
An. other	–	578	–	–
Non-mosquito	–	344	–	–

within a few hours of collection. Table 1 summarizes the number of specimens per species class by verification method and imaging platform. The 791 PCR-verified An. stephensi VectorCam specimens were drawn from two sources: 398 specimens provided by the University of Notre Dame laboratory (Dr. Neil F. Lobo) and 393 from Ethiopian field collections.

Handheld smartphone imaging

A supplementary handheld dataset was collected using smartphones paired with clip-on macro lens attachments, using both 10X and 15X magnification across three phone models: Samsung A15, Redmi 13C, and Moto G Play (2021). These models were selected to reflect phones commonly used across Africa based on cost and availability, in order to build diversity into the collected dataset. Handheld imaging was conducted at the University of Notre Dame because specimens contributed by the Notre Dame laboratory were already located there for PCR verification; imaging was performed on these specimens prior to PCR processing.

The handheld smartphone dataset comprised 1,049 images in total (Table 1). The 755 PCR-verified specimens represent the same physical specimens as those contributed by the University of Notre Dame laboratory to the VectorCam dataset, re-imaged using handheld smartphones.

Pretraining dataset

Pretraining of the classification models was conducted using a dataset of over 60,000 labeled images of field-collected mosquito specimens from Uganda, including *Anopheles* (comprising *An. gambiae* s.l., *An. funestus* s.l., and other *Anopheles* species), *Culex*, and *Mansonia* species, as part of prior VectorCam validation work [10]. The majority of specimens in this dataset were identified by trained entomologists using morphological keys [14]. This large, high-quality dataset, captured under standardized VectorCam imaging conditions, provided consistent and controlled images that were crucial for initializing model weights and building a robust foundation of morphological feature representations prior to fine-tuning on *An. stephensi*-specific field data from Ethiopia.

Data augmentation

To enhance model generalization and robustness across the variable conditions that are characteristic of real field environments, several augmentation techniques were applied during training. These included random horizontal and vertical flips, rotations, perspective transformations, color jitter, and image resizing, which together simulated realistic variability in mosquito positioning, lighting conditions, and image framing. Z-score normalization was also applied to standardize pixel values across datasets, ensuring consistency during training across both the VectorCam and handheld image sources.

Model architecture and training

This study employed a two-stage deep learning pipeline for both imaging modalities: an object detection model to locate and isolate individual mosquito specimens within each image, followed by a species classification model to assign a species identity to the detected specimen. For VectorCam images, the device provided a standardized optical configuration, imaging distance, magnification, and controlled illumination. Specimen orientation and position within the imaging field were intentionally allowed to vary during image acquisition to increase positional diversity in the training dataset and improve model robustness to variation expected during future field implementation. For the handheld pipeline, the same two-stage approach was applied under more variable background, lighting, camera distance, and specimen positioning conditions. Grad-CAM analysis was applied post-hoc to examine whether the models had learned to attend to morphological features such as wing venation, body shape, and leg patterning that are used in conventional entomological species identification.

Object detection using YOLOv5

For the handheld imaging pipeline, YOLOv5 [18] was used to detect and localize individual mosquito specimens within images prior to species classification. This object detection step efficiently isolated individual mosquitoes from complex and variable backgrounds, generating cropped bounding boxes that were then passed as input to the downstream species classification model. YOLOv5 was selected for its balance of rapid inference speed and accurate object detection, making it well suited for real-time processing in field environments where computational resources may be limited.

Species classification using EfficientNet-B1

Species classification was performed using EfficientNet-B1 [19] across all three classifiers developed in this study. EfficientNet-B1 is a CNN architecture that scales model depth, width, and input resolution in a coordinated manner to achieve high classification accuracy at a low computational cost. The model occupies 25.4 MB on device, making it suitable for deployment on smartphones and other resource-constrained hardware. This property was a consideration in model selection for this study, as both the VectorCam Android application and the handheld web application are intended for use in field settings where processing capacity and storage may be limited. EfficientNet-B1's pretraining on a dataset of over 60,000

labeled mosquito images provided a foundation of morphological feature representations across multiple species, with further fine-tuning applied to specialize the model for *An. stephensi* identification within the East African context.

Three distinct classifiers were developed and validated to address different surveillance scenarios and varying levels of available imaging equipment.

1. Multiclass VectorCam Classifier:

The VectorCam multiclass classifier was trained on VectorCam images to distinguish between seven classes: *An. stephensi*, *An. funestus* s.l., *An. gambiae* s.l., *An. other*, *Culex*, *Mansonia*, and non-mosquito. This classifier could enable comprehensive species-level surveillance at sentinel sites in East Africa equipped with VectorCam devices, providing simultaneous identification of all major co-occurring mosquito species alongside *An. stephensi*, and including a non-mosquito class to handle cases where non-target specimens are inadvertently imaged.

2. Binary VectorCam Classifier:

The VectorCam binary classifier was trained on VectorCam images specifically to distinguish *An. stephensi* from all other species. By narrowing the classification task to a single binary decision, this classifier is optimized for high positive predictive value in *An. stephensi* confirmation, which represents the most operationally critical use case for field surveillance programs responding to the presence of this invasive species.

3. Binary Handheld Smartphone Classifier:

The smartphone binary classifier was trained on images captured using operator handheld smartphones paired with clip-on macro lenses to distinguish *An. stephensi* from other species under variable and less controlled imaging conditions.

Training protocol

All three classifiers were developed using a multi-phase training approach. Pretraining was first conducted on the Uganda VectorCam dataset to establish morphological feature representations across a broad range of mosquito species. Transfer learning was then applied to specialize each pretrained model for the specific classification task at hand, utilizing the VectorCam Ethiopia dataset for the VectorCam classifiers and the handheld dataset for the smartphone-based classifier. This two-stage approach was used to mitigate the possible effects of a small training dataset for the primary classification task.

For each classifier, the dataset was first split at the specimen level into a training pool (80%) and a held-out test set (20%), stratified by species. Five-fold cross-validation was then performed on the training pool (70% train/10% validation per fold), with the held-out 20% test set reserved exclusively for final ensemble evaluation and never used during training or hyperparameter selection.

All models were trained using an input resolution of 1024×1024 for a maximum of 80 epochs, utilizing early stopping with a patience of 10 epochs. Optimization was performed using the AdamW algorithm with a learning rate of $1 \times 10^{-}4$ and a weight decay of $1 \times 10^{-}4$ and a batch size of 4 across all experiments. All training procedures were executed using GPU acceleration provisioned via Google Colaboratory. Data splits were performed at the specimen level, ensuring that all images derived from the same physical specimen were assigned to the same split and that no specimen appeared across multiple sets. The full pipeline is shown schematically in Fig. 3.

The top panel shows the inference pipeline. A mosquito specimen is captured using either a VectorCam device or a smartphone with clip-on macro lens. Individual specimens are localized using YOLOv5 [18] object detection, and the cropped specimen image is passed to the EfficientNet-B1 [19] classification model to generate a species prediction. The middle panel shows the system workflow from data collection through to field deployment and species identification output. The bottom panel shows the training pipeline. EfficientNet-B1 was pretrained on over 60,000 labeled images of mosquitoes from various sources, then fine-tuned on Ethiopia and Uganda field specimens across three classifier tasks: VectorCam multiclass, VectorCam binary, and smartphone binary. Training was performed using 5-fold cross-validation with majority voting ensemble predictions. Final performance metrics were evaluated per classifier.

The following analytical methods were applied to all three classifiers. Gradient-weighted Class Activation Mapping (Grad-CAM) was applied post-hoc to visualize the specific image regions that contributed to classification decisions, providing an interpretability analysis of model behavior.

Uniform Manifold Approximation and Projection (UMAP) was applied as a dimensionality reduction technique to visualize the learned feature space of each model. For one fold per classifier, penultimate-layer feature representations were extracted and projected into two dimensions, enabling visual inspection of class separability and clustering structure in the test data. UMAP projections were generated using n_neighbors=15, min_dist=0.3, and Euclidean distance metric.

To generate final predictions, a majority voting ensemble strategy was employed across the five cross-validation

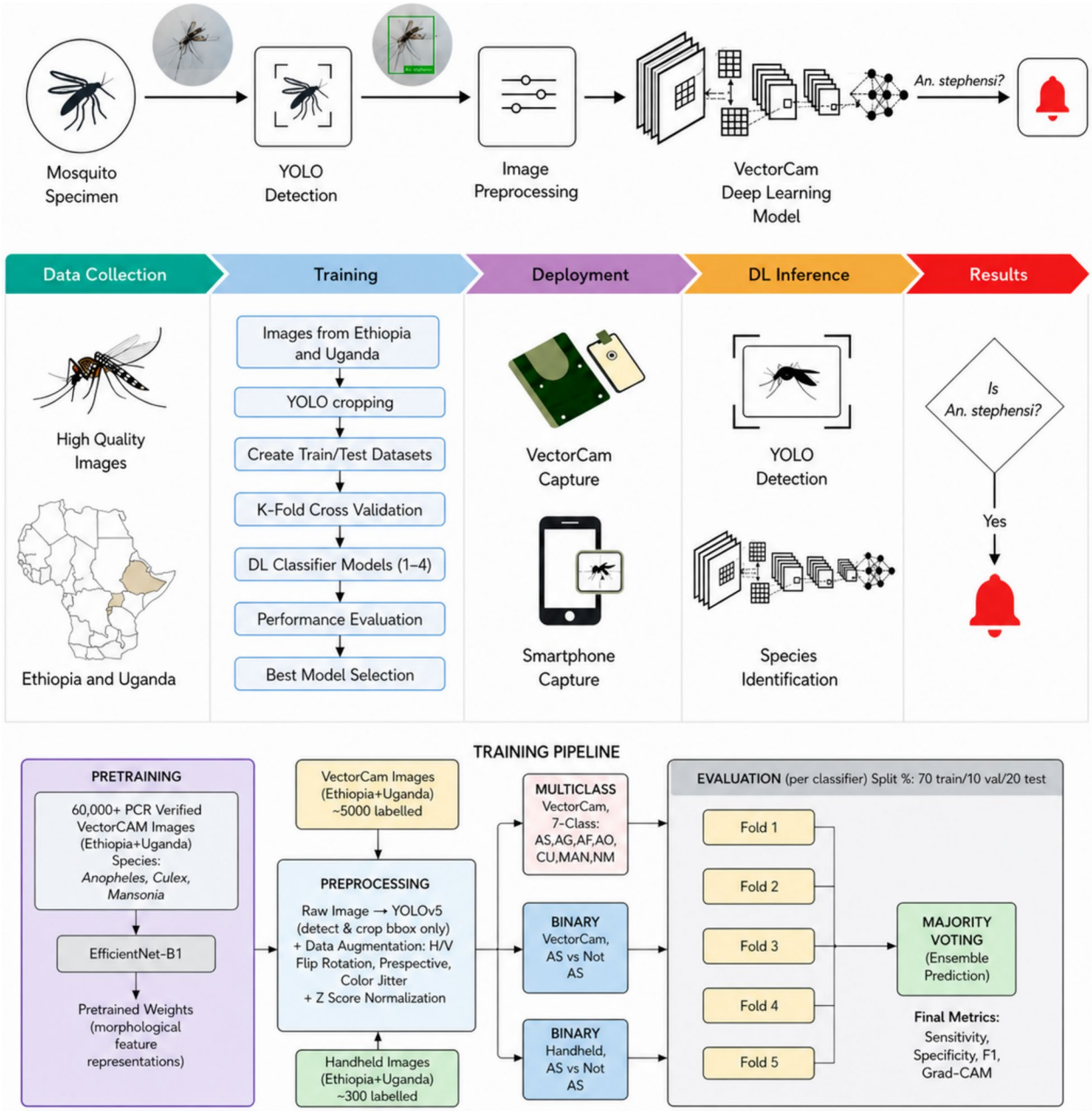


Fig. 3 Schematic of the *Anopheles stephensi* detection system pipeline across both imaging modalities

folds. For each test specimen, the predicted class from each of the five trained models was collected, and the final prediction was assigned as the class receiving the most votes. This ensemble approach provides predictions that account for variability across individual fold models, reducing the influence of fold-specific variability on final performance estimates.

All classification results reported below represent ensemble performance across 5-fold cross-validation using majority voting, unless otherwise stated. Performance metrics including positive predictive value, sensitivity, overall accuracy, and specificity were derived from the ensemble confusion matrices. All error values represent standard deviation across folds unless stated otherwise.

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manuscript text. All scientific content, data, analyses, and conclusions were generated and verified by the authors.

Results

All performance metrics reported in this section were calculated against ground-truth species identifications verified by PCR (*An. stephensi*, *An. gambiae* s.l., *An. funestus* s.l., *Culex*, and *Mansonia* specimens) or morphological identification by trained entomologists (*An. other* and non-mosquito specimens, and a subset of *An. gambiae* s.l. and *An. funestus* s.l. specimens), as described in the Methods.

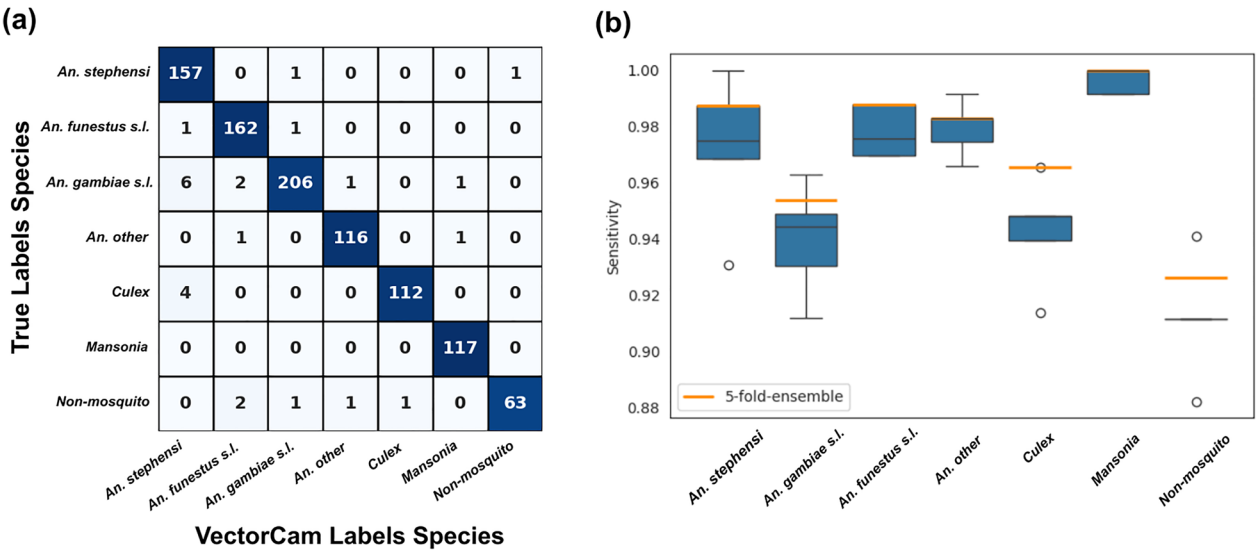
Multiclass classification with VectorCam

For *An. stephensi*, the classifier achieved a majority-vote sensitivity (recall) of 98.74%, with a positive predictive (precision) value of 93.45% and an overall accuracy (F-1 macro average) of 96.02% (Fig. 4, panel c). The sensitivity and positive predictive value confusion matrices are shown in panel (a), showing the distribution of correct and incorrect identifications across all seven classes.

Examining performance across individual folds, the box plot of per-fold sensitivity is shown in panel (b). 2.78% of *An. gambiae* s.l. specimens were misclassified into the *An. stephensi* class in the majority-vote ensemble matrix. The non-mosquito class showed the greatest fold-to-fold variability of any class, with an ensemble-level sensitivity of 92.65%. Summary performance metrics including overall accuracy, sensitivity, specificity, and positive predictive value for all seven classes are presented in panel (c), with all classes achieving greater than 95% sensitivity and specificity under the majority-vote ensemble. All ± values represent standard deviation across folds.

Binary classification with VectorCam device

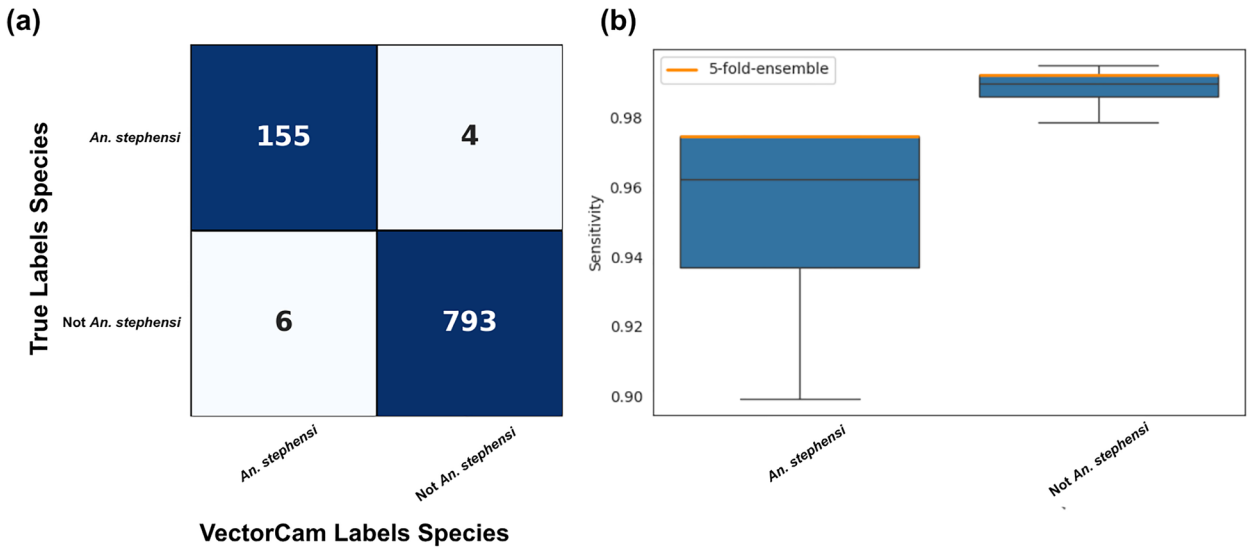
The VectorCam binary classifier was tasked with distinguishing *An. stephensi* from all other species (Not *An. stephensi)* using VectorCam images. Across 5-fold cross-validation, the model achieved a mean *An. stephensi* sensitivity of 97.48% and a Not *An. stephensi* sensitivity of 99.25%, with a misclassification rate for Not *An. stephensi* specimens of 0.75% (Fig. 5, panel a). The 5-fold ensemble metrics showed an *An. stephensi* positive predictive value of 96.27%, sensitivity of 97.48%, and a specificity of 99.25% (Fig. 5, panel c).



(c)

Class	Sensitivity	Specificity	Positive Predictive Value	Overall Accuracy	Support
An. stephensi	98.74	98.62	93.45	96.02	159
An. funestus s.l.	98.78	99.37	97.01	97.89	164
An. gambiae s.l.	95.37	99.60	98.56	96.94	216
An. other	98.31	99.76	98.31	98.31	118
Culex	96.55	99.88	99.12	97.82	116
Mansonia	100.0	99.76	98.32	99.15	117
Not-mosquito	92.65	99.89	98.44	95.45	68

Fig. 4 Ensemble performance of the VectorCam multiclass classifier across seven mosquito species classes



(c)

Class	Sensitivity	Specificity	Positive Predictive Value	Overall Accuracy	Support
An. stephensi	97.48	99.25	96.27	96.88	159
Not *An. stephensi*	99.25	97.48	99.50	99.37	799

Fig. 5 Ensemble performance of the VectorCam binary classifier distinguishing *An. stephensi* from all other species

The box plot of per-fold sensitivity illustrates the contrasting consistency between the two classes. The Not *An. stephensi* sensitivity showed an interquartile range of approximately 0.988 to 0.990 across folds. *Anopheles stephensi* sensitivity showed an interquartile range of approximately 0.940 to 0.970, with a minimum fold sensitivity of approximately 0.900 (Fig. 5, panel b). The sample size for *An. stephensi* was n=159 versus n=799 for Not *An. stephens*i. Summary performance metrics including overall accuracy, sensitivity, and specificity for both classes are presented in panel (c). All ± values represent standard deviation across folds.

It should be noted that a subset of Not *An. stephensi* specimens in this classifier were verified by morphological identification rather than PCR. As morphological identification carries an inherent error rate, particularly for species sharing overlapping morphological characteristics, some apparent classifier errors may reflect uncertainty in the ground-truth labels rather than true model misclassification. The reported sensitivity and specificity values should be interpreted with this in mind.

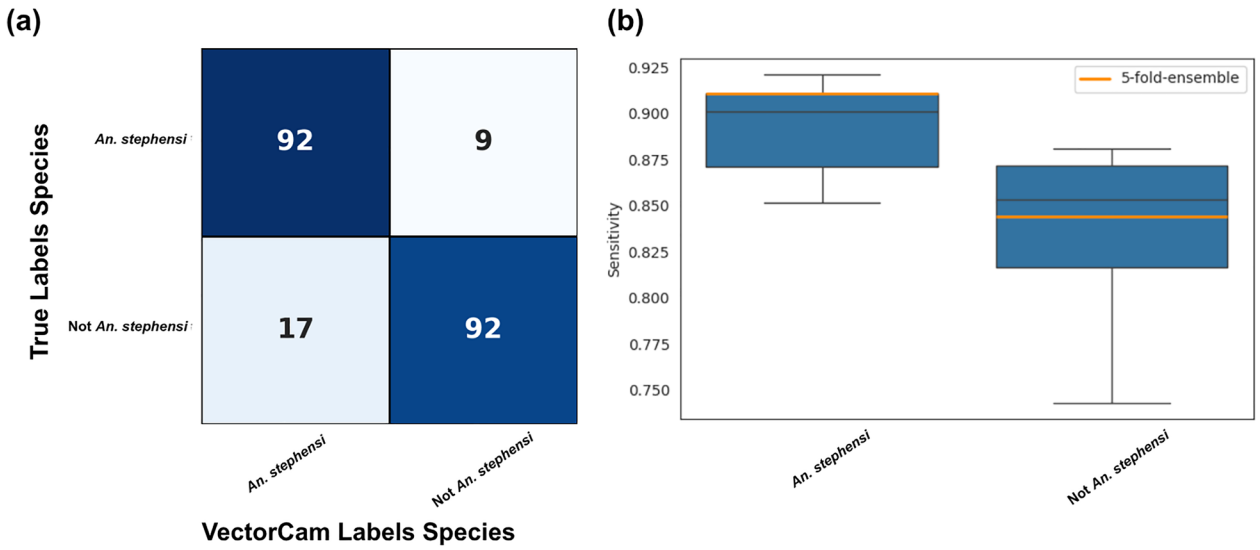
Binary classification with handheld device

The smartphone binary classifier was designed to distinguish *An. stephensi* from non-*An. stephensi* specimens (Not *An. stephensi*) using images captured via smartphones with clip-on macro lens attachments at the University of Notre Dame laboratory under ambient lighting conditions. Across 5-fold cross-validation, the ensemble classifier achieved a mean *An. stephensi* sensitivity of 89.11% (± 2.58%) and a positive predictive value of 84.4%, with an overall accuracy of 87.62% (Fig. 6, panel c). In the majority-vote ensemble confusion matrix, *An. stephensi* sensitivity was 91.09% and Not *An. stephensi* sensitivity was 84.40%, with 16% of Not *An. stephensi* specimens misclassified as *An. stephensi* and 9% of *An. stephensi* specimens misclassified as Not *An. stephensi* (Fig. 6, panel a).

The box plot of per-fold sensitivity shows that *An. stephensi* sensitivity showed a median of approximately 0.90 and an interquartile range of 0.878 to 0.910 across folds. The Not *An. stephensi* sensitivity showed a median of approximately 0.848 and a minimum of 0.750 across folds (Fig. 6, panel b). The Not *An. stephensi* class encompasses multiple species imaged under variable ambient conditions. The sample size for *An. stephensi* was n=101 versus n=109 for Not *An. stephens*i. Summary performance metrics are presented in panel (c). All ± values represent standard deviation across folds.

gradient-weighted class activation mapping

Grad-CAM visualizations were generated for one fold of the VectorCam multiclass classifier to examine the image regions most influential in driving classification



(c)

Class	Sensitivity	Specificity	Positive Predictive Value	Overall Accuracy	Support
An. stephensi	91.09	84.40	84.40	87.62	101
Not *An. stephensi*	84.40	91.09	91.09	87.62	109

Fig. 6 Ensemble performance of the handheld smartphone binary classifier distinguishing *An. stephensi* from all other species

decisions across all seven species classes (Fig. 7). Across initial layers, attention was distributed broadly across the mosquito body, reflecting the extraction of general morphological features. In the final layers, activation became progressively more localized, with the strongest responses concentrating on the wing venation patterns and body regions for most species. For *An. stephensi* and *An. funestus* s.l., high-intensity activation in the final layers was consistently localized to the wings, while for *Mansonia* activation spanned both the wings and thorax, which is consistent with the distinctive wing scaling that characterizes this genus. For the non-mosquito class, final layer activation concentrated on body shape and appendage structure, indicating that the model learned to attend to features that differentiate non-mosquito specimens from Anophelines at a structural level. The progressive sharpening of activation from initial to final layers across all classes provides evidence that the classifier is learning hierarchical, biologically meaningful morphological features rather than responding to background elements or imaging artifacts that would not be expected to generalize across field conditions.

Representative images are shown (Fig. 7) for each classification category, alongside Grad-CAM visualizations from earlier and final model layers. Warmer regions indicate image areas contributing more strongly to model predictions. Despite variation in specimen position and orientation, final-layer attention was generally concentrated on mosquito morphological regions, including the wings, legs, thorax, abdomen, and other body structures relevant to conventional entomological identification.

Uniform manifold approximation and projection

UMAP projections were generated using n_neighbors=15, min_dist=0.3, and Euclidean distance metric (see Methods). UMAP visualizations of the learned feature spaces confirmed that all three classifiers developed well-separated class representations in the test data (Fig. 8). In the VectorCam multiclass model, each of the seven species classes formed distinct clusters in the projected feature space, indicating that the model had learned discriminative representations for each class (Fig. 8a). Notably, the *An. stephensi* and *An. gambiae* s.l. clusters appeared in closer proximity to each other than to other species classes, consistent with the known morphological similarity between these two species and the pattern of misclassification observed in the confusion matrices. In the VectorCam binary model, the *An. stephensi* cluster was well separated from the Not *An. stephensi* cluster, though the Not *An. stephensi* points were spread across a wider region of the feature space, reflecting the heterogeneous nature of this class which encompasses multiple morphologically distinct species (Fig. 8b). A similar pattern was observed in the handheld binary

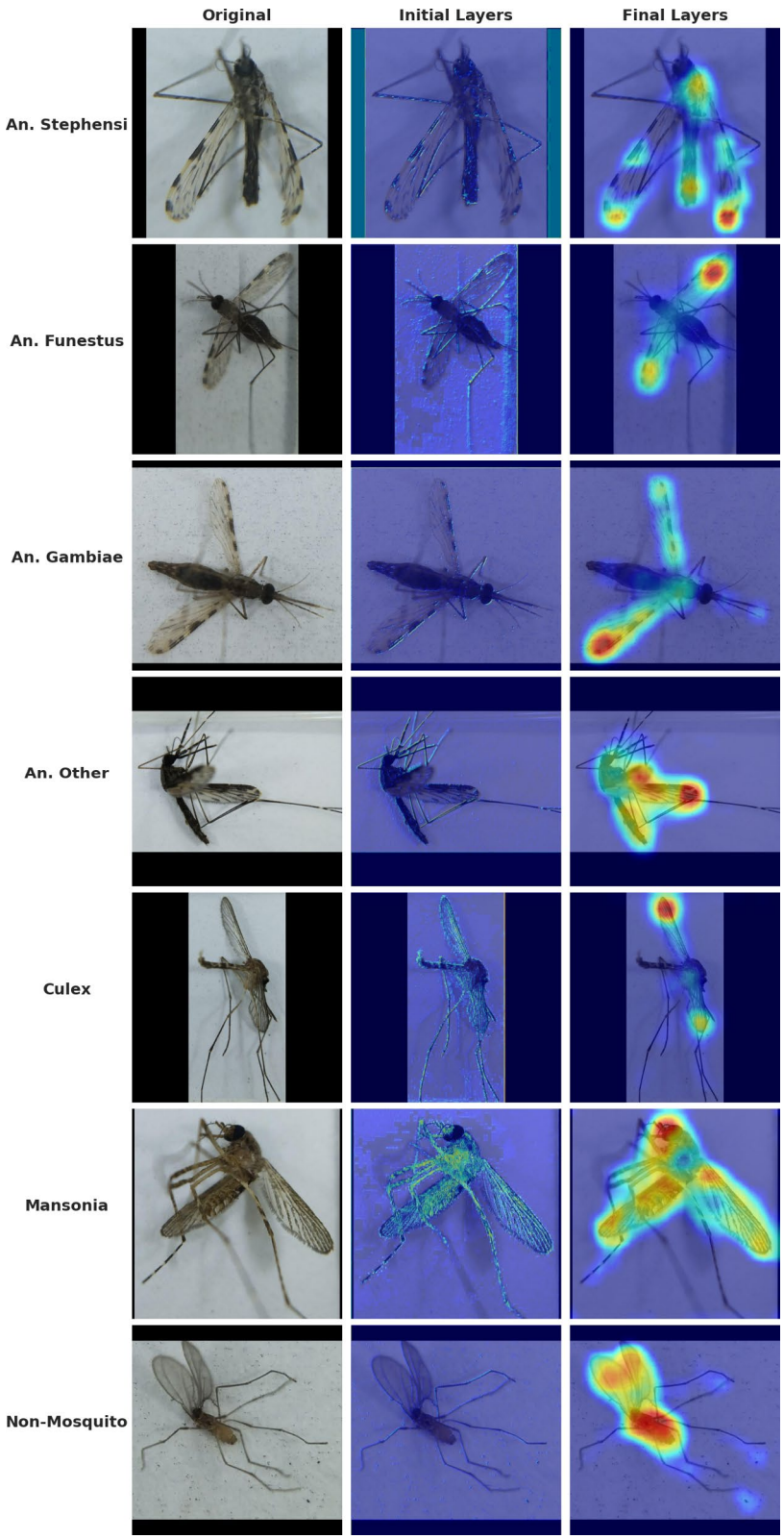


Fig. 7 Grad-CAM visualizations

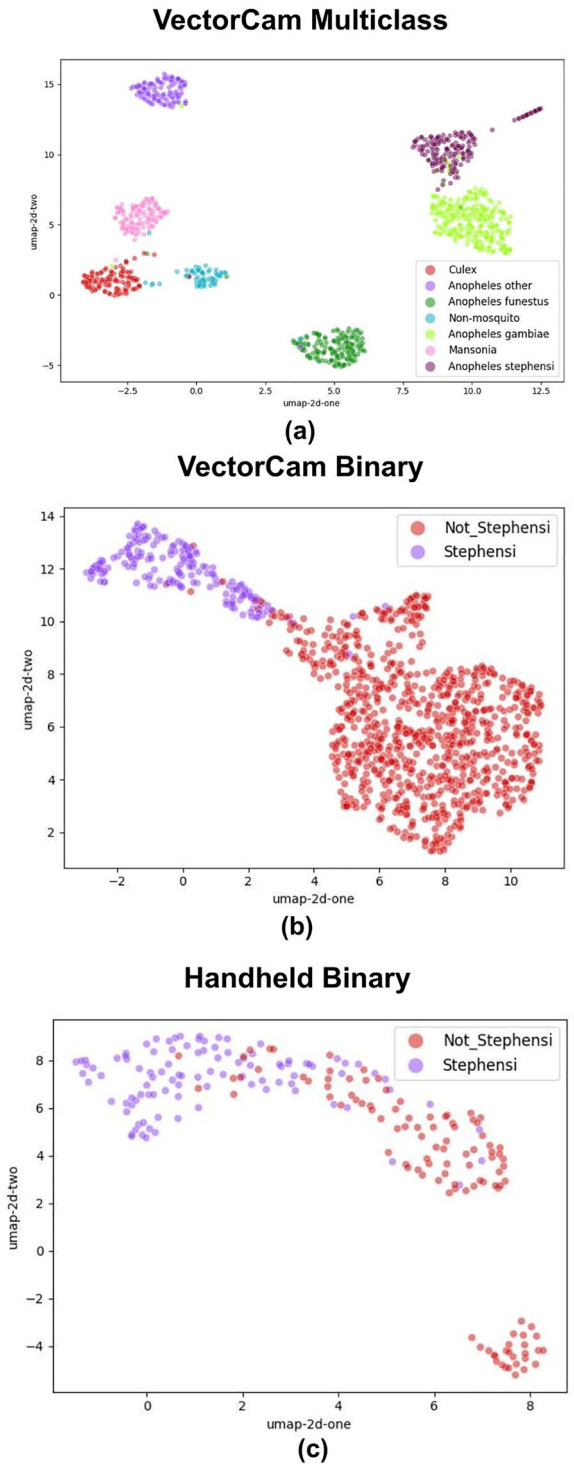


Fig. 8 UMAP visualizations

model, where *An. stephensi* and Not *An. stephensi* points formed largely separate regions, though with greater overlap than in the VectorCam binary model, consistent with the lower overall classification performance of the handheld classifier (Fig. 8c).

Discussion

This study developed and validated three deep learning classifiers for field detection of *An. stephensi* from photographic images, using VectorCam as the primary imaging platform and a handheld smartphone with clip-on macro lens as a supplementary approach. Specimens were collected from sentinel sites across Ethiopia and Uganda and molecularly or morphologically verified for species identity. An EfficientNet-B1 architecture was fine-tuned using transfer learning from a pretraining dataset of over 60,000 labeled mosquito images. The VectorCam multiclass classifier distinguished seven species classes and achieved an *An. stephensi* sensitivity (recall) of 98.74% and *An. stephensi* PPV (precision) of 93.45%. The VectorCam binary classifier distinguished *An. stephensi* from all other species and achieved a sensitivity of 97.48% and a specificity of 99.25%. The smartphone binary classifier achieved an *An. stephensi* sensitivity of 91.09% and a specificity of 84.4%.

The specificity of 99.25% achieved by the VectorCam binary classifier indicates that the rate of false positive *An. stephensi* identifications is low, which is relevant for field surveillance programs where false positive alerts could require confirmatory investigation. The standardized optical setup, controlled illumination, and fixed magnification of the VectorCam device reduce inter-image variability, allowing the classification model to operate on consistent image inputs across different operators and field sites.

The misclassification of 2.78% of *An. gambiae* s.l. specimens into the *An. stephensi* class in the multiclass model reflects the morphological similarity between these two species, which is the primary identification challenge in the East African context. The VectorCam binary classifier, which focuses specifically on the *An. stephensi* versus all other species discrimination, achieved an overall accuracy above 95% for both classes and may therefore be more appropriate for programs where *An. stephensi* confirmation is the primary objective, with the multiclass model providing broader species-level surveillance at sentinel sites.

The smartphone classifier was trained on approximately 1,200 images captured under variable ambient conditions, compared to the larger VectorCam dataset. The difference in sensitivity between the two modalities reflects the difference in training data volume and consistency rather than a limitation of smartphone imaging for this application. Collection of additional smartphone images across diverse field environments, varying in lighting conditions, phone models, geographic settings,

and operator experience, could be expected to reduce this gap in future iterations of the classifier.

The inclusion of a Non-mosquito class in the VectorCam multiclass model allows the classifier to handle cases where non-target specimens are inadvertently imaged. The class achieved an ensemble-level sensitivity of 92.65%, indicating that additional training data for this class may improve fold-level consistency in future iterations.

One limitation of this study is the difference between the datasets collected from Ethiopia and Uganda, which may introduce variation in model performance due to differences in environmental conditions and geographic morphological variation across mosquito populations. Expanding training datasets to include specimens from a broader range of African countries and ecological settings will be necessary for the classifiers to be applied reliably across the full geographic range where *An. stephensi* surveillance is needed. Establishing open repositories of labeled mosquito images captured under diverse field conditions could support this process and reduce duplication of data collection efforts across research groups.

Future development could consider the integration of uncertainty quantification methods, such as Monte Carlo dropout or ensemble confidence scoring, to allow field users to identify images where classification confidence falls below a defined threshold and flag these cases for review by a trained entomologist. Deployment of the VectorCam classifiers within the VectorCam Android application and the smartphone classifier via an open-access web application represent directions for future development that could extend *An. stephensi* detection capability to field programs across a range of resource settings.

Conclusions

This study demonstrates that accurate, field-deployable detection of *Anopheles stephensi* from photographic images is achievable using deep learning classification models trained on molecularly and morphologically verified field specimens. The VectorCam-based classifiers achieved *An. stephensi* sensitivity of up to 98.74% in the multiclass model and an overall accuracy of 96.88% and specificity of 99.25% in the binary model, indicating performance suitable for operational vector surveillance applications. A supplementary smartphone binary classifier achieved a majority-vote *An. stephensi* sensitivity of 91.09%, extending detection capability to programs that do not have access to dedicated VectorCam hardware. Grad-CAM analysis indicated that classification decisions were predominantly associated with wing venation patterns and leg morphology, consistent with the features used in conventional entomological identification. These classifiers represent a practical and scalable contribution to decentralized *An. stephensi* surveillance capacity in Africa, where timely and accurate detection of this invasive vector is critically needed to guide targeted malaria control responses as the species continues to expand its geographic range, reducing reliance on molecular verification or specialist taxonomic training for field-level species identification.

Abbreviations

AF	*Anopheles funestus*
AG	*Anopheles gambiae*
AI	Artificial intelligence
AO	*Anopheles other*
AS	*Anopheles stephensi*
CDC-LTC	Center for disease control light trap catch
CNN	Convolutional neural network
CU	*Culex*
Grad-CAM	Gradient-weighted class activation mapping
MAN	*Mansonia*
NM	Non-mosquito
PCR	Polymerase chain reaction
PPV	Positive predictive value
PSC	Pyrethrum spray catch
UMAP	Uniform manifold approximation and projection
YOLO	You only look once

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Author contributions

SA conceived the study and provided overall supervision. PM performed data preparation, developed and trained the classification models, and was the primary contributor to manuscript writing. ASK and AZ contributed to data preparation, model development, and manuscript writing. DL contributed to model development and provided guidance on the coding pipeline. NFL provided entomological expertise, contributed field-collected An. stephensi and An. gambiae s.l. specimens used in model training and validation, and oversaw PCR verification of specimens in his laboratory. DLo and JCu performed molecular verification of mosquito specimens at the University of Notre Dame. DY provided entomological expertise on An. stephensi and oversaw mosquito specimen collection across sentinel sites in Ethiopia. BLC and DZG collected field specimens at sentinel sites in Ethiopia, imaged specimens, performed and troubleshot PCR verification of the Ethiopian specimens, field tested the identification system and provided feedback that informed its development, and reviewed he manuscript. AS contributed to figure preparation and manuscript writing. SMP and MRT reviewed the manuscript. RC provided expertise in computer vision and AI. JMCa provided overall direction. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate.
Not applicable.

Consent for publication
Not applicable.

Competing interests
SA, MRT, and SP are co-founders of Vector Control Innovations, Inc (VCI), a 501(c)(3) non-profit organization that develops and distributes VectorCam. All other authors declare no competing interests.

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